Local Area Networks

- These are private communications networks:
  - They are used to interconnect the computing resources within an organisation.
- All devices attach to a shared transmission medium.
  - They can provide data rates up to 1 Gbps:
- The rationale for interconnecting devices:
  - Proliferation of computing devices,
  - Growth in multimedia applications,
  - Increased in client/server applications.

Local Area Networks

- LANs can be classified according to:
  - Their topology/layout e.g. ring, star, bus…,
  - The type of transmission medium used e.g. TP Cable, Wireless…
  - How they access the shared Tx medium (aka as Access Control) e.g. CSMA, Token…
- There are three broad types of LANs:
  - PC LANs (aka the Access Layer):
    - Used to connect low-cost devices e.g. PCs, small server devices etc.
    - Used in most organisations,
    - These LANs are cheap to implement and maintain and consequently their performance and speed is low.

Typical Applications for LANs

- Backend LANs (aka The Core Layer):
  - Used to connect high-cost devices e.g. Mainframes, Supercomputers, mass storage devices,
  - Mainly used for the bulk transfer of data,
  - Used in large organisations,
  - These LANs as costly to implement and maintain and consequently their performance and speed is high.

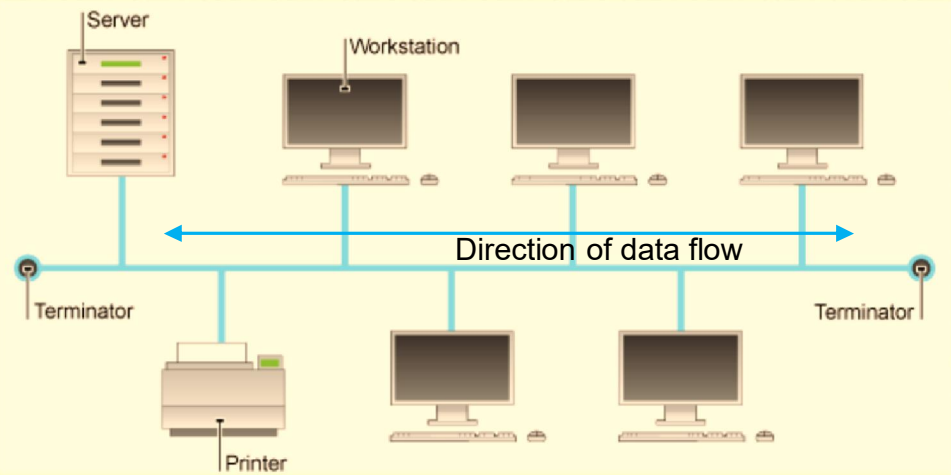
Typical Applications for LANs

- Backbone LANs (aka The Distribution Layer):
  - Used to provide connectivity between PC LANs and Backend LANs,
- The use of single LAN is not desirable because:
  - Reliability. If the single LAN develops a fault the effects can be devastating,
  - Capacity. More devices connecting to a LAN reduces its performance,
  - Cost. Connecting Low cost PC devices using the same type of LAN used to connect High cost server devices would be impractical.

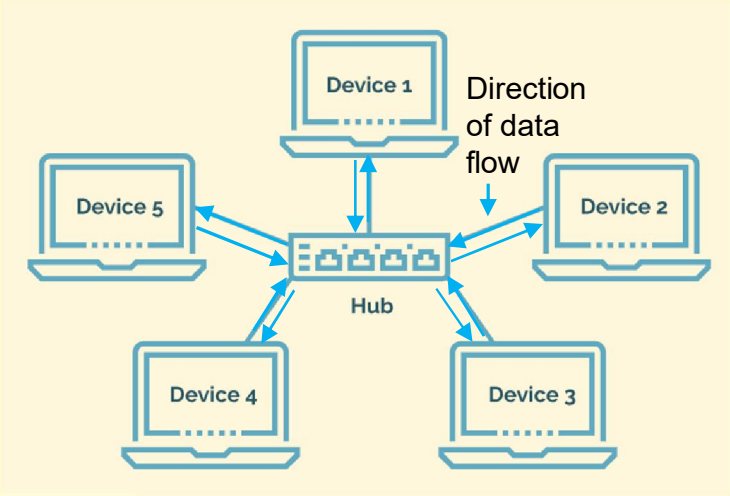
LAN Topologies

- There are three LAN topologies to consider: ***Bus***, ***Ring*** and ***Star.***
- *Bus* LAN characteristics:
  - A *linear* transmission medium is used.
  - Each station attaches to the *bus* using a tap.
  - There are no closed loops.
  - Transmissions travel in both directions i.e. *bi-directional*, and can be seen by <u>every</u> attached device.
  - The bus is terminated by a *terminator* device which removes the transmission signal.
  - Example of a bus LAN is *Ethernet.*

LAN Topologies

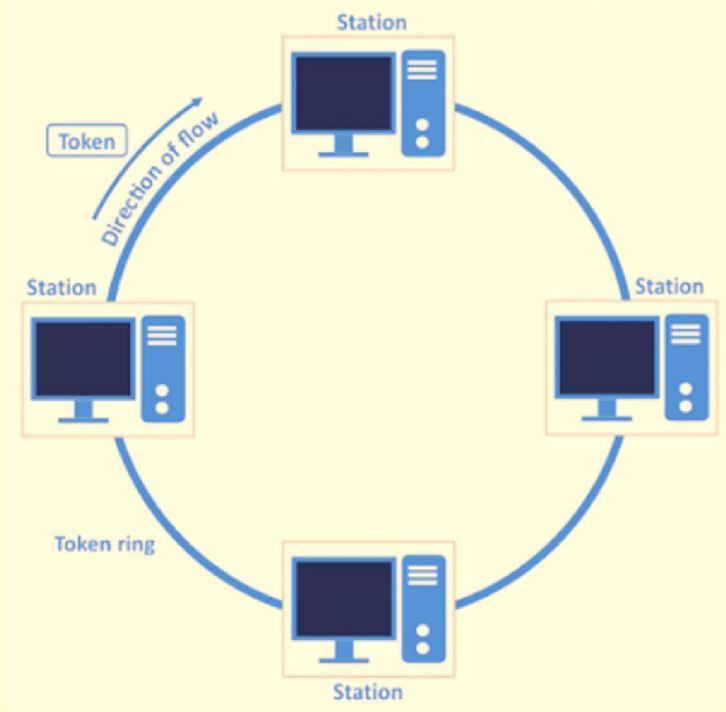


Bus LAN



Star LAN

Ring LAN



LAN Topologies – contd.

- *Ring* LAN characteristics:
  - Consists of a set of interconnected *repeaters* (using point-to-point links) in a *ring* configuration.
  - Repeaters receive data on one link and transmit them *bit-by-bit* onto another link.
  - The links are *uni-directional.*
  - Each station attaches at a ***Repeater***.
  - The transmission signal must be <u>physically</u> removed from the network.
  - Example of a ring LAN is the *IBM Token Ring.*

LAN Topologies – contd.

- *Star* LAN characteristics:
  - Consists of a central *hub* component of which there are two types:
    - *Shared-medium* hub.
    - *Switched-LAN* hub.
  - Each station is attached by two links (*transmit* and *receive*).
  - All transmissions travel via the Hub device.
  - Example of a star LAN is an ATM.

LAN Topologies – contd.

- Common characteristics of LANs:
  - Data is transmitted in small blocks known as *frames*
    - Each frame contains Data plus Control Information (similar to HDLC).
  - Stations must gain access to the shared transmission medium:
    - Access must be provided on a fair and equitable basis i.e. there must be some controlling mechanism employed.